BOYUE ZHANG

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Education

- **Zhejiang University** *Sep. 2023 - Jun. 2027 (Expected)*
  B.S. in Biological Sciences, Bioinformatics track, GPA: 3.65/4.0 Hangzhou, China
  - **Coursework:** Computational Biology (98/100), Biophysics (96/100), Bioinformatics (97/100)
  - **Honors:** Professional Excellence Scholarship, Academic Excellence Model
- **The Chinese University of Hong Kong** *Sep 2026 - Dec 2026 (Upcoming)*
  Exchange Student, International Asian Studies Programme (IASP) Hong Kong, China

Projects

- **Walter and Eliza Hall Institute (WEHI) InSPIRE Research Intern, Fully Funded** *Jun 2026 - Present*
  Tools: Flow Cytometry, Degranulation Assay, HEK293T Culture, GROMACS, Rosetta, ESM2 [ ] [ ]
  - *Supervised by Dr. Emma Petley; PIs: Prof. Matt Call & A/Prof. Melissa Call, WEHI InSPIRE Program*
  - **Design:** Redesigned the CAR-T costimulatory (4-1BBζ) TMD into its natural trimeric form to test whether restored geometry improves T cell activation (lab's proCAR3 hypothesis)
  - **Wet Lab (Functional Validation):** Optimized 4 TMD variants; ran flow cytometry and degranulation assays against 2 tumor lines, confirming surface expression (73–74%) and activation; benchmarking vs. conventional 4-1BBζ ongoing
  - **Dry Lab (Computational Screening):** Ran a four-route, 120+ μs study (all-atom MD, umbrella sampling/SMD, MARTINI 3 self-assembly, insertion PMF) to separate assembly from membrane insertion as causes of expression differences; all 4 Rosetta/ESM2-scored designs converged to stable, symmetric trimer interfaces
- **iGEM 2025 (world's largest synthetic biology competition): Technical Lead** *Aug 2024 - Oct 2025*
  Tools: CRISPR-i/a, Optogenetics, Confocal Microscopy, Python, R, GROMACS, AlphaFold3 [ ] [ ] [ ]
  - *Advised by Prof. Ming Chen and Assoc. Prof. Fan Yang, Zhejiang University*
  - **Leadership:** Led ZJU-China's 14-member interdisciplinary team *LumaManta* (wet/dry lab, hardware, software) to a Silver Medal, building an all-in-one device for real-time water pollutant and eDNA monitoring
  - **Biosensor design:** Engineered *BiChromaLogic*, a light-controlled CRISPR logic circuit converting 2 light inputs into 4 sensor proteins on demand (**169%** activation, **92%** repression)
- **Computational Design of CrtW and CrtZ for Enhanced Astaxanthin Biosynthesis** *Sep 2025 - Present*
  Tools: Rosetta, GROMACS, ESM2, PyMOL, AlphaFold3, HPLC [ ] [ ]
  - *Advised by Dr. Yangwei Jiang, Qiushi Chair Prof. Ruhong Zhou's Lab, Zhejiang University*
  - **Design (268 candidates):** Engineered CrtW (β-carotene ketolase), rate-limiting in astaxanthin biosynthesis, screening 320 residues via structure-based (Rosetta FuncLib) and sequence-based (ESM2, ThermoMPNN) design
  - **Simulation (8 μs MD):** Evaluated the top 6 candidates; G142P+I192L gave the most extended substrate positioning and most stable binding pocket, outperforming static predictions with no stability loss
  - **Validation (+46.7% yield):** Confirmed the top mutant in vitro over wild-type
- **Sex Dimorphism in DEN-Induced Hepatocellular Carcinoma** *Mar 2026 - Present*
  Tools: R (DESeq2, WGCNA), Python, Untargeted Metabolomics, Transcriptomics [ ]
  - *Advised by Dr. Wenbin Zhou, Prof. Meihua Sui's Lab, ZJU School of Medicine*
  - **Multi-omics curation (174 samples):** Curated 9,082 raw spectral features into a 468-metabolite matrix from paired serum metabolomics and liver transcriptomics
  - **Confounder-controlled screen (20 markers):** Modeled sex $\times$ disease interaction with tumor burden as covariate to separate sex effects from male tumor load; 4 orthogonal evidence lines narrowed $\sim$190 candidates to 20 (permutation $p = 0.001$), 13 unreported
  - **Estrogen (ERα) axis:** Confirmed the direction of the metabolic reprogramming axis with a bidirectional intervention arm; panel queued for targeted MS/MS validation

- **Provincial Research Grant (SRTP): A Sustainable Pigment Biosynthesis System** *March 2025 - Feb 2026*
  Tools: GROMACS, LC-MS, SnapGene, GSMM, Python, R, Optogenetics [🔗] [🌐]
  - *Advised by Prof. Ming Chen, Zhejiang University*
  - **Enzyme & metabolic engineering:** Identified a vioE mutant (+32% activity) via semi-rational design and MD simulation, and two gene knockouts (+48%/+23% yield) via genome-scale metabolic modeling
  - **Genetic circuit design:** Built dual orthogonal quorum-sensing circuits and a blue-light-inducible split-Cre system for programmable, reversible pigment switching

Honors and Awards

- **WEHI InSPIRE Scholarship** *Melbourne 2026*
  Walter and Eliza Hall Institute of Medical Research — 8,000 AUD; competitively selected, ZJU-nominated
- **iGEM 2025 Silver Medal, Grand Jamboree** *Paris Oct 2025*
  International Genetically Engineered Machine Competition, world's largest in synthetic biology — Technical Lead [🌐]
- **Grand Prize (Champion), National Synthetic Biology Innovation Competition** *Aug 2025*
  Chinese Society for Biotechnology & Shenzhen Institute of Synthetic Biology — one of China's largest in synthetic biology
- **Gold Medal, 18th "Dandelion" Undergraduate Innovation Competition** *Mar 2026*
  Zhejiang University — CTO, LumaManta eDNA-based aquatic pathogen detection
- **First Prize (Champion), 2nd "Forbel Cup" National Pet Industry Innovation Competition** *Jan 2025*
  Shanghai Society of Animal Husbandry & Veterinary Medicine — Led "Pet Smart Health" team
- **Provincial SRTP Research Grant** *2025*
  Zhejiang Provincial Department of Science and Technology — 12,000 RMB
- **5th Place, Stand-Up Paddleboard 200m, University Sports Championship** *Jun 2025*
  Zhejiang University
- **Academic Honors & Scholarships**
  Zhejiang University 2023–2025
  - Professional Excellence Scholarship (2023–2024)
  - Third-Class Scholarship (2024–2025)
  - Innovation & Entrepreneurship Model; Academic Excellence Model (2024–2025)

Skills

- **Programming Languages:** Python, R, Bash/Shell, C
- **Computational Biology & Bioinformatics:** Rosetta, AlphaFold3, GROMACS, ESM2, ThermoMPNN, PyMOL, SnapGene, CHARMM-GUI, MEGA
- **Data science & omics:** Transcriptomics, metabolomics, WGCNA, DESeq2, TCGA, GEO, scikit-learn, pandas, NumPy, SciPy
- **Molecular & Synthetic Biology:** CRISPR-i/a, Optogenetics, Gibson Assembly, Molecular Cloning, Site-Directed Mutagenesis, GSMM
- **Cell Biology & Protein:** Human Primary T Cell / Jurkat / HEK293T Culture, Viral Production & Purification, Flow Cytometry, Degranulation Assay, Cell Counting, Protein Expression & Purification, Western Blot, Immunofluorescence, Confocal Microscopy
- **Languages:** English (IELTS 6.5)

Leadership and Volunteer Experience

- **Advisor, ZJU-China iGEM 2026 Team** *2026*
  Zhejiang University — advising the incoming team on project design and computational modeling [🌐]
- **Class Monitor** *Sep 2023 - Jul 2025*
  School of Life Sciences & College of Yunfeng, Zhejiang University [🌐]
  - Led 30+ students across two cohorts, organizing 10+ career seminars and coordinating peer mentoring for 15+ students
  - Managed volunteer programs including animal welfare, West Lake conservation, and Hangzhou Zoo docent training